System), IBM's first DBMS. In this model each record has information in parent/ child relationship like a tree structure. The collection of records is called as record types, which are equivalent to tables in relational model. The individual records are equal to rows. See Figure 3.1

In the above model we have many advantages like less redundant data, efficient search, data integrity and security. This model also has few limitations like complex to implement and difficulty in handling many to many relationships.

3.2.2 Network model

The first developed network data model was IDS (Integrated Data Store) at Honeywell. Network model is similar to Hierarchical model except that in this model each member can have more than one owner. The many to many relationships are handled in a better way. This model identified the three database components Network schema, Sub schema and Language for data management. See Figure 3.2

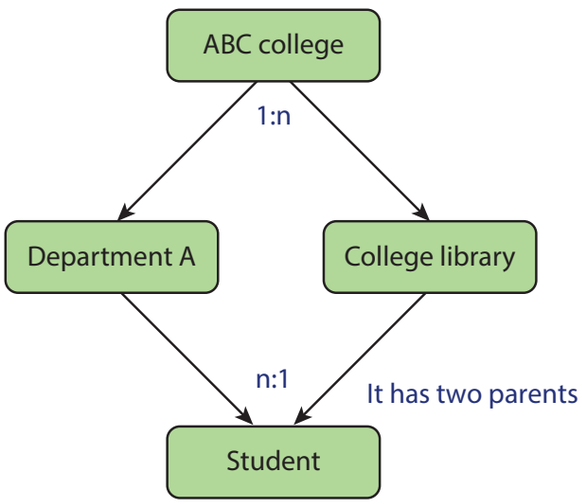


Figure: 3.2 Network Model

Network schema – schema defines all about the structure of the database.

Sub schema – control on views of the database for the user

Language – basic procede for accessing the database.

The major advantage of this model is the ability to handle more relationship types, easy data access, data integrity and independence. The limitation of network model is difficulty in design and maintenance.

3.2.3 Relational model

Oracle and DB2 are few commercial relational models in use. Relational model is defined with two terminologies Instance and Schema. See Figure 3.3

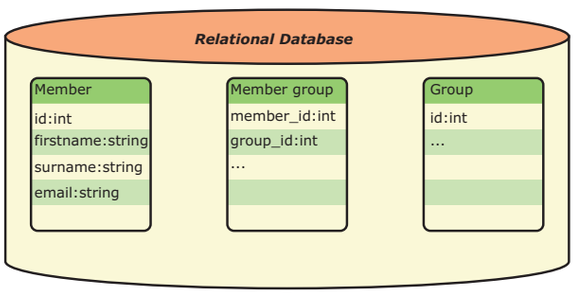


Figure: 3.3 Relational database model

Instance – A table consisting of rows and columns

Schema – Specifies the structure including name and type of each column.

A relation (table) consists of unique attributes (columns) and tuples (rows).

3.2.4 Object-oriented database model

This model incorporates the combination of Object Oriented Programming(OOP's) concepts and database technologies. Practically, this model serves as the base of Relational model. Object oriented model

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